Processing of Big Educational Data in the Cloud Using Apache Hadoop

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Abstract—**The amount of available educational data is growing and the educational institutions try to organize and understand large complex datasets. Since these big data are an emerging field of study, there is no simple solution to process or analyze them. This paper aims to provide a model approach to process big educational data from the Moodle system in the cloud using the Apache Hadoop cluster. An experimental implementation of the proposed model approach is included to validate this solution. The results then suggest that the proposed model approach has a potential to be widely used for the processing of big educational data, especially with user interaction data.**

Keywords-Big educational data; Cloud computing; Moodle; Apache Hadoop

I. INTRODUCTION

Nowadays, educational institutions need to have access to relevant information in order to provide quality education with improved efficiency and also minimal cost to their students. Furthermore, most of the educational institutions store massive amounts of data both in structured and unstructured formats. Due to the persistent growth and increasing availability of these data, it is needed to propose new models to process them.

During the past few years there has been a growing interest in the term "big data", fostered by the broad availability of various data acquisition, storing and processing platforms [1, 2]. Big data make it possible to mine learning information for insights regarding student performance, behavior and learning approaches such as evaluation of a much wider range of student actions, such as how long they devote to readings, what electronic sources they use, and how quickly they obtain the information they want. Observing the way students perform in the e-learning system is a way to assess the efficiency of this system, which creates various data types such as log files, and help stakeholders to improve instructional materials and the whole educational process in its entirety. Despite generating increasing volumes of educational data, until recently, only a few institutions have been able to exploit the wealth of information they collect [1, 3, 4, 5, 6, 7].

Thus, authors are focused to propose and validate a model approach, which provides the basic methodology to process big educational data in the cloud with the use of the Apache Hadoop cluster. This paper is arranged in the following way: in the next section, background of study including importance of education and educational data, the nature of big data, cloud computing and related platforms, tools and services will be described. In the section 3, a problem statement and tools used are briefly described. Section 4 details the proposed model approach and its validation. Finally, related results, discussion and conclusions are outlined.

II. RELATED WORK AND BACKGROUND

The use of Internet in education has created a new context known as e-learning and web-based education in which large amounts of data about teaching and learning interaction are endlessly generated and ubiquitously available [6]. E-learning courses may be then provided through learning management systems such as Moodle, Blackboard and Sakai, or learning platforms such as Knewton and DreamBox. In this so-called e-learning ecosystem, the amount of e-learning log data grows exponentially, introducing the variety and complexity of data analysis by using different educational data mining approaches and techniques [3, 7].

These educational data can be categorized to five different categories [3]: the first one pertaining to student identity and on boarding (identity data), and then four student activity-based datasets that have the potential to improve learning outcomes (user interaction data, inferred content data, system-wide data, and inferred student data). Big educational data should help to improve learning performance of students, enhance working effectiveness of teachers and reducing administrative workload [1, 4]. Furthermore, two specific fields that are significant to the exploitation of big data in education are educational data mining and learning analytics [3]. Especially, student learning performance prediction, individual assessment, curriculum and courses recommendation, learning patterns analysis and also learning personalization are some of the key aspects [1]. Thus, a robust, scalable and practical architecture to process these data is urgently needed [7]. According to Fernandez et al. [4], a cloud platform is a natural structure for the implementation of data mining techniques and their application to growing big educational data. For example, Hashem et al. [8] described the use of cloud computing in big data as follows: "*Large data sources from the cloud and Web are stored in a distributed fault-tolerant database and processed through a programing*

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model for large datasets with a parallel distributed algorithm in a cluster."

Aghabozorgi et al. [3] focused on the challenges and analysis with big educational data involved with uncovering or extracting knowledge from large datasets and they emphasized the use of different educational data mining approaches, tools and techniques to mine these vast amounts of data. Cen et al. [1] introduced some new opportunities for big data analytics to improve the efficiency of students' learning and maximize their knowledge retention. They proposed data-driven learning analytics for identification of the optimal patterns of learning that could advise on the most effective format, style and pace of learning for the individual student.

In general, there can be identified two main ways to process big educational data [9]. The first one requires some hardware costs and investments or at least configuration of the computer cluster. This can be done by installing Apache Hadoop, an open-source platform written in Java for reliable, scalable and distributed computing, under the UNIX type operating system. The second option is to save all the hardware costs and use some of the open-source or commercial platforms and cloud resources. Both of these solutions have the similar base. The replicated File System (FS), which handles all the parallel and distributed processing operations automatically. It can be a job distribution, optimization, error handling, thread access and many more complicated tasks [2, 9].

Zheng et al. [7] then proposed a logging architecture to support the whole lifecycle of these data. It is composed of five modules: log collection, transport, storage, computation and service. Michalik et al. [10] introduced a design of big data architecture in the education system. The lowest layer covers data sources. These data can be stored in traditional SQL databases or NoSQL databases. In the next layer is Apache Hadoop. This system processes the huge amount of data from the lowest layer. The main advantage of this platform is the fact that it can work with NoSQL data and also relation data. At the same level is Apache Hive. A data warehouse is in the middle layer. The highest layer is only for creating useful analysis from obtained data. These analyses are output of the process of the possible change in the education system at universities. Marjanovic et al. [5] then proposed a big data infrastructure deployed as Hadoop platform in order to improve education process. The platform is convenient for integration with the learning management system Moodle platform. Their model for managing big data in educational institutions enables manipulation of data generated through users' online activities. Output of the manipulation process is then used to improve the quality of e-learning. Hashem et al. [8] then discussed the relationship between big data and cloud computing, big data storage systems and Apache Hadoop technology.

III. Problem Statement and Tools Used

The primary objective of this paper is to offer a solution and introduce a model for the distributed processing of big educational data in the cloud using Apache Hadoop. For the purpose of this objective the new model approach is proposed and validated.

The partial objectives are intended to meet the following objectives. The first one is to conduct a literature review that accounts for earlier research in the area and explores the impact of big data, distributed processing and cloud computing on education, including technology-related terms. The second one is to compose a suitable architecture and choose the right technologies for the deployment and validation of the proposed model approach. And finally, process selected big educational data and then present the results to show the feasibility of this solution.

The processing tasks use the MapReduce framework [11], which is here represented by its open-source implementation Apache Hadoop [2]. Computing and data infrastructure was deployed with the use of the open-source OpenStack software platform, which is designed to deploy and manage large networks of virtual machines, as a highly available and highly scalable Infrastructure as a Service (IaaS) cloud computing platform. Ubuntu Server 14.04, Java 8 and Hadoop 2.7.2 were used for the deployment of the Apache Hadoop cluster in the cloud. The other platforms, tools and services used are then described more in detail in the following section.

The validation of this approach will then highlight some of the potential insights that are available from big educational data. These data used are user interaction data, i.e., engagement metrics, click rate, page views and other activities, time and date, etc., as defined by Aghabozorgi et al. [3], and Zheng et al. [7], i.e., big log analysis for e-learning, and were received from the Moodle system of the University of Pardubice.

IV. Model Description and Validation in the Cloud

The model on the Fig. 1 consists of the most important parts described in the previous section. An input is represented by big educational data from an e-learning system. In this case, only terminal, i.e., user operations, environment preferences, webpage, etc., and also application, i.e., operation logs, user assessment, examination content, forum discussion, experiment results, etc., layer data are considered. For more information, see e.g. Zheng et al. [7]. Furthermore, big educational data are collected through API or other interfaces and transported to the raw data storage with the use of the most suitable platform, tool or service. Data storage and data processing are performed in the cloud. Raw data storage system then stores historical data completely, securely and permanently for future data mining and analyzing. Data-intensive computing framework is applied to analyze massive raw data to dig out valuable information. A final output is in this model represented by the result data storage.

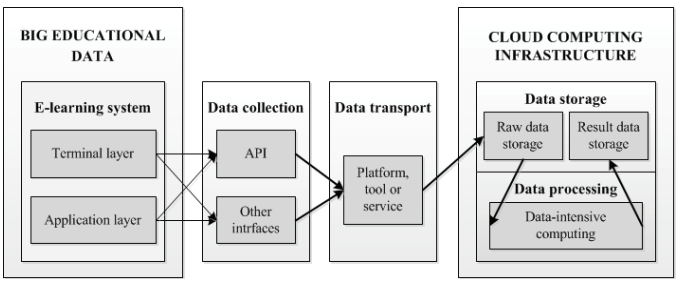

Figure 1. A model approach to process big educational data in the cloud

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According to the characteristics and model analyzed above, a testing architecture is designed as shown in the Fig. 2. The architecture can be also easily constructed on the basis of open-source platforms, tools and services. The API was used to limit programming only to the computational tasks and transfer data from the Moodle system to the cloud. It is an interface of a service, that is usually reachable by login information and it provides the possibility to outsource a specific activity in the cloud. Its creation is covered in the work of Henning [12]. The experimental implementation of the proposed model approach is done with the use of the following platforms: Apache Flume, Apache Hadoop and Hadoop Distributed File System (HDFS), Apache HBase, Apache Hive, Apache Sqoop and OpenStack. This so-called Hadoop ecosystem can contain various numbers of platforms, tools and also services. They are improving the performance of Apache Hadoop and help programmers and data analysts to make their work more efficient. Most of them come from Apache Foundation and usually work only with the Apache Hadoop platform. Every one of them has its specific functionality, and combining them with Apache Hadoop makes this ecosystem very powerful [5, 9].

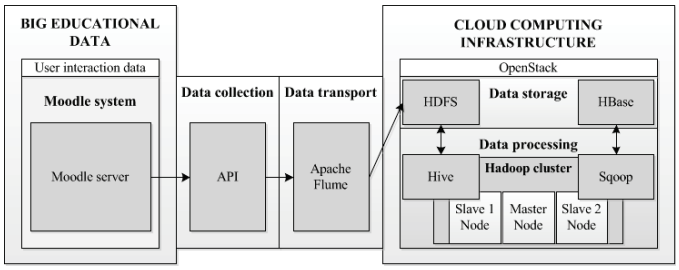

Figure 2. An experimental implementation of the proposed model approach

Apache Hadoop can easily run in a cloud environment [2], therefore, it was set up as a three node Hadoop cluster in the small laboratory conditions for testing purposes to show the feasibility of the newly proposed model approach. Firstly, user interaction data generated by users in the Moodle system are collected by collector interfaces, in this case by API, and then are transferred to HDFS by Flume, an open-source platform for log transport. Secondly, Hive, an open-source platform for batch processing to facilitate querying and managing large datasets residing in distributed storage, is used to analyze these data to obtain statistical analysis results, which are processed by Apache Hadoop and exported to HBase, an open-source, column-oriented database management system that runs on top of the HDFS. For this purpose, Apache Sqoop is utilized. OpenStack is used to deploy and manage a large number of virtual machines and the software-side infrastructure required to support it. Apache Hadoop cluster communicates with other components using TCP/IP protocol.

V. Results and Discussion

Information about all users in the Moodle system, i.e., for the role of the instructor, course editor, student, etc., and their interactions are contained in the log files, which were further processed. Therefore, interactions for all these users per day are shown in the Fig. 3, where the horizontal axis shows date and the vertical axis shows the total number of interactions. This curve reveals obvious periodicity, where the highest number of interactions is realized on Wednesdays. In addition, on the weekend of each week or holidays, the total number of interactions decreased significantly. This phenomenon may be resulted by many factors such as learning styles and strategies, curriculum plan, cognitive abilities into education or the other factors that affect the efficiency of the learning process. More information about this phenomenon can found in [6].

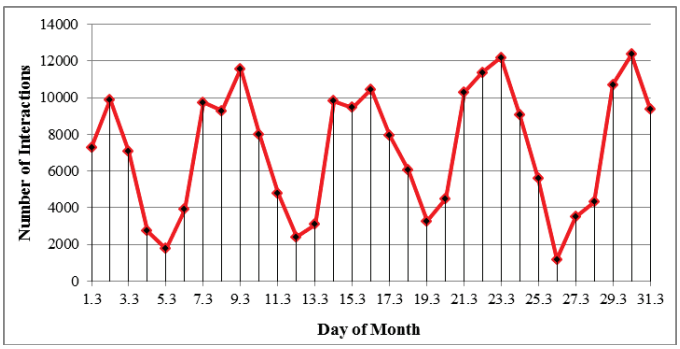

Figure 3. Number of users' daily interactions with the Moodle system

Numbers of selected users' interactions with the Moodle system are shown in the Table 1. As can be seen from these results, most of the users finish their work without logging out or their session may also expire. Course view it is the most common activity of users, followed by assignment view. It indicates that many users will start with browsing the course view first and then followed by assignment view. The third highest occurrence is user login. There are also quite a lot of login errors, almost 40 per day. Finally, forum add discussion activity is neglected by users very frequently.

TABLE I. Numbers of Selected Users' Interactions with the Moodle System

Interaction (activity)	Monthly	Daily (high/low)
User login	15607	503 (1070/82)
User logout	7126	230 (457/22)
Login error	1215	39 (66/8)
User view	1098	35 (63/2)
Course view	50133	1617 (4412/281)
Course enroll	302	10 (37/0)
Resource add	469	15 (23/0)
Resource update	341	11 (18/0)
Resource view	13785	445 (959/85)
Assignment add	674	22 (35/0)
Assignment update	1541	50 (68/0)
Assignment update grades	1466	47 (151/0)
Assignment upload	8201	265 (410/16)
Assignment view	31267	1009 (2547/144)
Forum view discussion	981	32 (106/5)
Forum add discussion	124	4 (13/0)
Wiki view	6327	204 (565/0)

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In contrast to similar studies, this paper utilizes the cloud solution to deploy the Apache Hadoop cluster. Furthermore, it clearly describes the platforms (modules) used and finally, it presents more detailed findings about users' interactions with the Moodle system than the other studies. The results obtained in this paper may be used to integrate within the existing in-house educational framework as used by institutions also as to keep pace with the rapid adoption of modern technologies for learning environment. Also, as stated by Fernandez et al. [4] cloud computing may help to solve the challenges with regard to optimizing resource computation in the institution, storage and communication requirements, big data processing, energy efficiency, and dealing with dynamic concurrency requests highlight the need of the use of a platform that meets scalable demands and cost control.

VI. Conclusions and Future Research

In this paper, authors offered an effective approach to analyze educational data from the Moodle system in the cloud using Apache Hadoop. The main objective and partial steps were completed. The growth of big educational data is parallel to growth of e-learning and also related systems. Therefore, the main contribution of this paper is the newly proposed model approach for processing big educational data generated from the Moodle system, which was also implemented and validated as an experimental architecture.

In the first part, the theoretical background of related terms was established focusing on the importance of big educational data for the educational institutions in the context of cloud computing. Based on these findings a new model approach was proposed and implemented as an experimental architecture. The second part of this paper was focused on the importance of the distributed approach and Apache Hadoop in the processing of big educational data. Finally, the most important results and findings were highlighted. The deployment of this solution can save the resources, especially costs, and get the results in the reasonable time, which is an important part in the decision-making process. It should also help to improve the related processes of e-learning.

Future research should be focused on more detail analysis of user interaction data such as was proposed by Zheng et al. [7] to cover the whole lifecycle of e-learning log data which includes log collection, transport, storage, computation and service. It will also involve more studies on applying the proposed model approach to analyze the learning patterns of users, and to evaluate learning performance in the related e-learning systems. Furthermore, more data sources should be identified in the education environment, i.e., not only from related e-learning systems, but also from other systems in the educational institutions, to fully use the advantage of the distributed data processing framework MapReduce, its open-source implementation Apache Hadoop and also the proposed model approach. Finally, one challenging question is how to extract meaningful information out of the vast click-stream data (big educational data streams), that is, the record of each click on course materials made by each student in the course. These pathways can then give teachers and instructors valuable information about the educational effectiveness of materials in a particular course.

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